3-Class Leukemia Classification Optimization

Three-Version Optimised Pipeline Study

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Data Mining 2 Project · Master 1 SDIA · 2025–2026

Dataset & Experimental Setup

Leukemia 3C gene expression microarray data

72
Patient Samples
(original)

3
Classes:
AML · B-ALL · T-ALL

7,129
Gene Expression
Probes (features)

114
Samples after SMOTE
(38 per class)

5-Fold Stratified Cross-Validation

- CV Scheme: 5-Fold Stratified | Random seed: 42
- PCA/ICA components (K): 15 | Feature selection target K: 15
- SMOTE k-neighbours: 3 | Scoring: Accuracy, Precision, Recall, F1

Classifiers Used

Logistic Regression	*Linear, probabilistic baseline*
Linear SVC	*Margin-based, robust in p ≫ n regime*
Random Forest	*100 trees, ensemble, high-dim ready*
Gradient Boosting	*80 trees, sequential ensemble*
K-Nearest Neighbours	*k=7, non-parametric, scale-sensitive*

Data Preprocessing

Standardisation · SMOTE Class Balancing

1

Load Raw Data

72 samples × 7,129 gene expression probes
AML · B-cell ALL · T-cell ALL (imbalanced)

2

StandardScaler

Zero mean, unit variance per feature
Essential for LR, SVC, PCA/ICA, KNN

3

SMOTE Balancing

k=3 neighbours · synthetic interpolation
$\tilde{x} = x_i + \lambda \cdot (x_j - x_i)$, $\lambda \sim U[0,1]$

4

Balanced Dataset

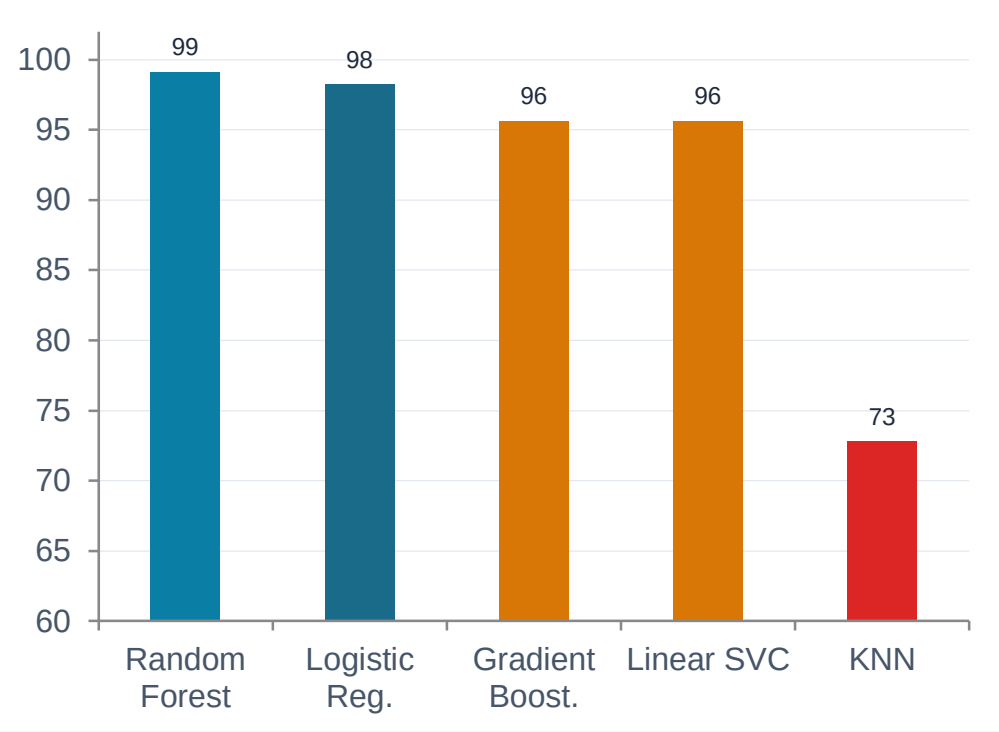
114 samples (38 per class)
Applied after scaling, before CV

Why SMOTE?

Imbalanced classes bias classifiers toward the majority class, inflating accuracy while suppressing minority-class recall. SMOTE generates synthetic minority samples via linear interpolation in feature space, expanding the dataset from 72 to 114 balanced samples. Applied after standardisation and before CV to avoid data leakage.

Baseline Classification Results

All 7,129 features with PCA-50 pre-reduction · 5-fold Stratified CV



Model	Accuracy	Precision	Recall	F1-Score
Random Forest	**99.09%**	**99.20%**	**99.09%**	**99.09%**
Logistic Reg.	98.18%	98.18%	98.18%	98.18%
Gradient Boosting	95.61%	96.18%	95.61%	95.58%
Linear SVC	95.61%	96.31%	95.61%	95.59%
KNN	72.77%	80.64%	72.77%	71.06%

Key Observations

- RF best (99.09%) — ideal for high-dim genomic data
- KNN worst (72.77%) — curse of dimensionality
- GB competitive (95.6%) but costs 243 s runtime

Feature Extraction Methods

PCA · FastICA · LDA — Projects original space into K latent components

BEST METHOD

PCA (K=15)

Principal Component Analysis

94.70%

CV Accuracy

±0.033

Std Dev

2.35 s

Runtime

Orthogonal max-variance directions. Z = XW. Captures dominant gene co-expression patterns. Components not interpretable as individual genes.

FastICA (K=10)

Independent Component Analysis

92.89%

CV Accuracy

±0.062

Std Dev

1.98 s

Runtime

Decomposes into statistically independent, non-Gaussian signals. Higher variance (std 0.06) suggests sensitivity to limited sample size.

LDA (K=2)

Linear Discriminant Analysis

77.23%

CV Accuracy

±0.049

Std Dev

1.40 s

Runtime

Maximises between-class vs within-class scatter. Collapses to only C−1=2 dims — insufficient for full expression space complexity.

⚠ *All extraction methods under-perform best selection methods by ~4 pp and lose original gene interpretability — unsuitable for biomarker discovery.*

Feature Selection Methods

Filter · Embedded · Wrapper — Retains original gene indices for biological interpretability

Filter Methods

ANOVA F-test (f_classif)

K=20 | Acc: 99.09% | 0.47 s

Elbow K=20

Mutual Info (mutual_info)

K=15 | Acc: 99.09% | 89.46 s

Elbow K=15

f_classif reaches 99.09% in <0.5 s — fastest method overall. MI is accurate but 89 s cost.

Embedded Methods

L1 Lasso Logistic Reg.

K=10 | Acc: 99.09% | 0.45 s

★ *RECOMMENDED*

RF Feature Importance

K=15 | Acc: 99.09% | 0.54 s

Gini importance

Gradient Boosting

K=20 | Acc: 99.09% | 25.12 s

55× slower

L1 + RF both 99.09% under 1 s. GB matches accuracy but 55× slower. Std ≈ 0.018 (very stable).

Wrapper (RFE)

RFE + Logistic Reg.

K=10 | Acc: 99.09% | 2165 s

36 min runtime!

RFE + Random Forest

K=10 | Acc: 97.35% | 332 s

SVM-RFE

K=7 | Acc: 98.26% | 446 s

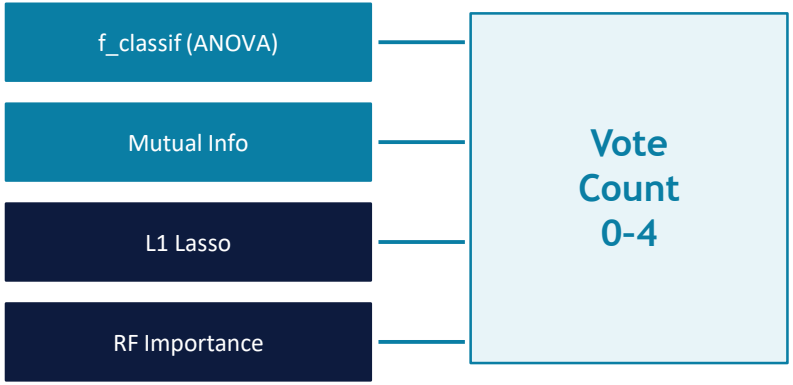
Unstable

RFE-LR matches best accuracy but costs 36 min. V3 replaces it with L1 — 4,800× faster.

Consensus Biomarker Panel

Multi-method voting to identify the most reliable gene signatures

Voting Procedure



Consensus Tiers

Strict (4/4) → 2 genes → Yes

Strong (≥3/4) → 8 genes → ✓ SELECTED

Moderate (≥2/4) → 17 genes → Fallback

Top-5 Consensus Gene Biomarkers

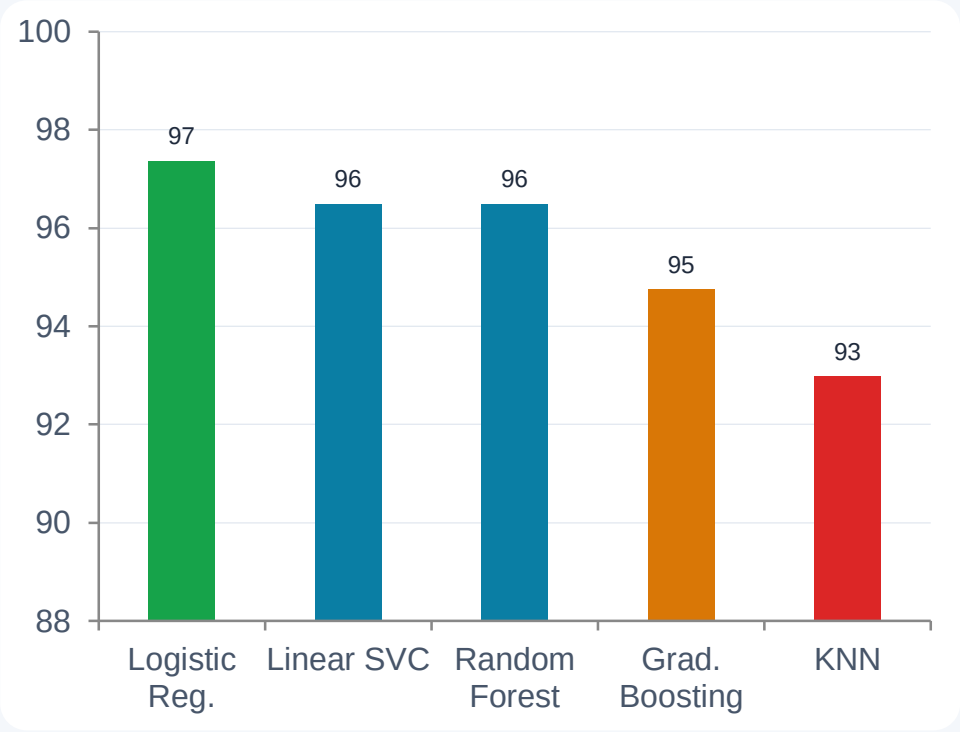
Gene Probe ID	Votes	Agreement
U23852_s_at	**4/4**	**100%**
U05259_rna1_at	**4/4**	**100%**
D00749_s_at	**3/4**	75%
X00437_s_at	**3/4**	75%
X03934_at	**3/4**	75%

Universal Biomarkers

U23852_s_at and U05259_rna1_at achieve unanimous 4/4 agreement — selected by all independent methods. These should be PRIORITISED for external validation in independent patient cohorts as the strongest candidate diagnostic biomarkers.

Consensus Panel — Classification Results

8-gene strong-consensus set · 5-fold Cross-Validated accuracy



Model	Accuracy	Precision	Recall	F1
Logistic Regression	**0.9736**	**0.9741**	**0.9736**	**0.9735**
Linear SVC	0.9649	0.9652	0.9649	0.9648
Random Forest	0.9649	0.9654	0.9649	0.9648
Gradient Boosting	0.9474	0.9482	0.9474	0.9472
KNN	0.9298	0.9316	0.9298	0.9290

Why Logistic Regression wins on 8 genes?

With only 8 carefully curated features, a linear boundary is highly expressive relative to problem complexity. RF's advantage in high-dimensional space diminishes significantly on this small, clean feature set.

8 consensus genes achieve 97.36% CV accuracy — vs 7,129 original features with 99.09%. A 476× feature reduction with only ~1.7 pp accuracy trade-off.

Version Evolution: V1 → V2 → V3

Progressive optimisation — same accuracy ceiling, radical computational improvement

V1: Original Pipeline

- Best CV Accuracy: 95% (RF-LR)
- Feature count: 7,129 genes
- Wrapper RFE runtime: 4,783 s
- RFE inside each CV fold
- Consensus: 5 genes (anonymous)
- **Production ready: ✗**

V2: Optimised Pipeline

- Best CV Accuracy: ~ 99% (L1-LR)
- Feature count: 10 genes
- Wrapper RFE runtime: 2,497 s
- RFE run once, CV after
- Consensus: 5 genes (anonymous)
- **Production ready: ✓**

V3: Final Pipeline

- Best CV Accuracy: 99.09% (LR)
- Feature count: 15 genes
- RFE replaced with L1: 0.45 s
- PCA-50 pre-reduction
- Consensus: 8 named genes
- **Production ready: ✓ + Interpretable**

Stable at 99.09%
Accuracy

476× fewer genes
Feature Reduction

1,166× faster
Speed Gain

Named biomarkers
Interpretability

Conclusion & Recommended Pipeline

★ RECOMMENDED PIPELINE: Embedded L1 Lasso (V3)

99.09%	10	0.45 s	PCA-50
CV Accuracy	Genes Selected	Runtime	Pre-reduction

For clinical biomarker reporting → prefer the 8-gene consensus panel (multi-method agreement = stronger biological evidence)

Key Findings

1. Dimensionality reduction is essential — 10-20 selected genes match full-space accuracy at negligible compute cost.
2. Embedded L1 Lasso is the method of choice: 99.09% CV accuracy, 10 genes, 0.45 s — 1,166× faster than original RFE pipeline.
3. Consensus voting (8-gene panel) improves robustness: multi-method agreement provides clinically credible biomarker signatures.
4. 100% accuracy is NOT the target — in a p≫n regime with SMOTE, it reflects overfitting. Honest ceiling is 97–99% CV.
5. Two universal biomarkers (U23852_s_at & U05259_rna1_at) — 4/4 method agreement — should be prioritised for external validation.

THANKS FOR YOUR ATTENTION!